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FACULTY OF COMPUTING AND INFORMATICS
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FINAL YEAR PROJECT - 1

**Intelligent Knowledge Management System
for a Research Institution**

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Table of Contents

	Catalog	
Abstract		V
Acronyms	VII	Error! Bookmark not defined.
Chapter One :Introduction		1
1. Introduction		1
1.1. Background of the Organization		1
1.2. Background of the Project		2
1.3. Motivation		2
1.4. Statement of the Problem		4
1.5. Objective of the Study		5
1.6. Feasibility Analysis		6
1.7. Scope of the Project		8
1.8. Significance of the Project		10
1.9. Target Beneficiaries		11
1.10. Methodology		12
1.11. Limitation of the Project		14
1.12. Team Composition		15
Chapter Two: Requirement Analysis and Specification		16
2.1. Introduction of Existing System		16
2.2. Proposed System		17
2.3. Requirements		18
2.4. System Models (SRS)		22
Actors:		22
Chapter Three: System Design		36

3.1. Introduction	36
3.2. Purpose of System Design	36
3.3. Design Goals	36
3.4. Deployment Diagram	37
3.5. Architectural Design	39
3.6. User Interface (UI) Design	41
3.7. Database Design	42
3.8. Machine Learning Algorithm Design	47
3.9. Summary	50

List of Figures

Figure 2.1: Use Case Diagram for the IKMS.....	18
Figure 2.2: Activity Diagram for Document Upload and Processing.....	22
Figure 2.3: Sequence Diagram for a Filtered Search Request.....	23
Figure 2.4: System Class Diagram.....	23
Figure 2.5: UI Mockup of the Main Search and Filter Page.....	24
Figure 2.6: UI Mockup of the Search Results Page.....	24
Figure 2.7: UI Mockup of the Researcher Dashboard.....	24
Figure 2.8: UI Mockup of an Institution Profile Page.....	25
Figure 3.1: System Deployment Diagram.....	28
Figure 3.2: Entity-Relationship Diagram for the System Database.....	34

List of Tables

Table 1.1: Estimated Project Budget.....	5
Table 1.2: Project Team Composition.....	12
Table 2.1: Use Case for Saving a Search Query.....	19
Table 2.2: Use Case for Uploading a Document.....	20
Table 2.3: Use Case for Managing Institutions.....	21
Table 3.1: Users Table Schema.....	32
Table 3.2: Institutions Table Schema.....	32
Table 3.3: Authors Table Schema.....	32
Table 3.4: Documents Table Schema.....	33
Table 3.5: Document_Authors Table Schema.....	33
Table 3.6: SavedSearches Table Schema.....	33

Abstract

Research institutions are currently facing a significant challenge in managing the escalating volume of scholarly publications, datasets, and research materials. The inability to efficiently organize and retrieve this information leads to knowledge fragmentation and missed opportunities for innovation. This project details the design and specification of an "Intelligent Knowledge Management System," a web-based application designed to address these challenges. Unlike existing fragmented systems that merely redirect to external publisher sites, our platform provides a centralized, locally-hosted repository for Ethiopian research, ensuring direct and permanent access to scholarly works. The system will leverage Natural Language Processing (NLP) techniques including TF-IDF for keyword generation and Latent Dirichlet Allocation (LDA) for topic modeling to automate the organization of research documents. Key features include a domain-aware semantic search with synonym support, content-based recommendations, author and institution profiling, and a personalized user dashboard with saved search alerts. The system is designed with a 3-Tier Architecture, employing a Python-based backend, a PostgreSQL database for structured metadata, and an Elasticsearch engine for efficient text indexing. By creating a centralized and intelligent platform, the system aims to significantly enhance the efficiency of knowledge discovery, foster a more collaborative research environment, and ultimately improve research outcomes within the Ethiopian health sector.

Acronyms

1. AI: Artificial Intelligence
2. API - Application Programming Interface
3. ERD - Entity-Relationship Diagram
4. EPHI - Ethiopian Public Health Institute
5. FYP - Final Year Project
6. GUI - Graphical User Interface
7. LDA - Latent Dirichlet Allocation
8. ML - Machine Learning
9. NLP - Natural Language Processing
10. PDF - Portable Document Format
11. RDBMS - Relational Database Management System
12. SRS - Software Requirement Specification
13. TF-IDF - Term Frequency-Inverse Document Frequency
14. UI - User Interface
15. UML - Unified Modeling Language

Chapter One :Introduction

1. Introduction

This chapter introduces the Intelligent Knowledge Management System (IKMS) project, outlining its context, motivations, objectives, and significance within Ethiopia's research ecosystem. The chapter provides a comprehensive overview of the project's background, problem statement, scope, and methodology, establishing the foundation for the subsequent design and implementation phases.

1.1. Background of the Organization

Ethiopia's national health research ecosystem is driven by pivotal institutions such as the Ethiopian Public Health Institute (EPHI) and specialized academic centers like the Jimma University Center for Biomedical Research (JU-CBMP). These organizations are critical engines of knowledge production, generating a continuous stream of research findings, technical reports, and scholarly publications aimed at informing public health policy and addressing pressing national health challenges.

However, a significant "discoverability gap" impedes the full utilization of this valuable knowledge. Research outputs are often confined within isolated institutional repositories, disparate digital libraries, or individual hard drives. This fragmentation creates substantial barriers for researchers, policymakers, and students seeking to find, access, and synthesize existing evidence. The current paradigm often relies on basic digital catalogs or manual search methods, which lack the intelligence to understand semantic meaning, contextual relationships, or the specialized terminology of the health domain. Consequently, valuable insights remain siloed, leading to duplicated research efforts, missed opportunities for collaboration, and a slower translation of evidence into practice.

It is within this context that our project is positioned. The proposed system is conceived not as a simple informational website or a static digital library, but as a specialized knowledge discovery tool. Its purpose is to actively bridge this discoverability gap by intelligently processing, interlinking, and surfacing research content. By applying advanced Natural Language Processing (NLP) and machine learning, the system will

transform a collection of documents into an interactive and intelligent map of knowledge, enabling users to uncover hidden connections and relevant information that would otherwise remain obscured.

1.2. Background of the Project

In the modern academic and scientific landscape, research institutions serve as critical hubs of innovation, continuously generating vast and growing repositories of knowledge through research papers, datasets, and scholarly articles. However, this exponential growth in information presents a significant management challenge. Traditional knowledge management systems, often relying on manual indexing and basic keyword matching, have proven inadequate for handling the scale and complexity of contemporary research outputs. These limitations frequently result in valuable research being overlooked, duplication of efforts, and ultimately, a retardation of scientific progress and innovation.

The evolution of knowledge management has progressed from physical archives and card catalogs to digital databases and, more recently, to intelligent systems capable of understanding content meaning. This transformation has been largely driven by advancements in Artificial Intelligence, particularly in Natural Language Processing (NLP). NLP has emerged as a transformative technology that enables computers to understand, interpret, and manipulate human language. From early rule-based systems to modern statistical and deep learning approaches, NLP has matured to offer sophisticated capabilities including semantic analysis, topic modeling, and contextual understanding.

This project, the "Intelligent Knowledge Management System for Research Institutions," emerges from the convergence of these technological advancements with persistent institutional needs. Our approach builds upon established NLP methodologies like TF-IDF for keyword extraction and Latent Dirichlet Allocation (LDA) for topic modeling, combined with enhanced semantic search capabilities. Rather than pursuing complex, resource-intensive deep learning models, we focus on implementing robust, interpretable techniques that provide immediate practical value while remaining feasible within our project scope and timeline.

The system specifically addresses the challenges of health research documentation by incorporating medical terminology understanding through synonym mapping and automated metadata extraction from PDF files. By leveraging these NLP technologies, we aim to create a solution that not only stores information but truly understands research content, enabling more intelligent organization and significantly improving the efficiency and effectiveness of knowledge discovery for researchers and academic professionals.

1.2.1. Comparative Analysis with Existing Systems

Our proposed Intelligent Knowledge Management System represents a paradigm shift from existing approaches to research dissemination in Ethiopia. Unlike current institutional repositories that merely redirect users to external publisher sites (creating access barriers and broken links), our platform hosts research documents directly, ensuring permanent, free access for Ethiopian researchers. While international platforms like Elsevier and Springer offer advanced features, they operate behind expensive paywalls and lack focus on Ethiopian research contexts. Academic social networks like ResearchGate provide free access but are commercial platforms that aggregate links rather than host content, with algorithms optimized for global rather than Ethiopian research priorities.

What distinguishes our system is its multi-layered innovation: First, we solve Ethiopia's specific "redirection problem" by hosting content locally rather than linking externally. Second, we provide cross-institutional aggregation—unifying research from multiple Ethiopian institutions while preserving their individual identities. Third, our AI features are specifically trained on Ethiopian research corpora, enabling context-aware understanding that generic international systems cannot provide. Fourth, we ensure data sovereignty by keeping Ethiopian research within Ethiopia, unlike current systems where content is hosted on international servers. Finally, our platform offers unique analytics on Ethiopia's research landscape—a feature completely absent in existing systems.

This combination of direct hosting, Ethiopian context-awareness, cross-institutional unification, and data sovereignty creates a platform fundamentally different from and superior to any currently available solution for Ethiopian researchers.

1.3. Motivation

Our team's motivation for this project is deeply rooted in a shared passion for applying cutting-edge information science and artificial intelligence to solve tangible, real-world problems that impact our community. As students immersed in the academic environment, we have witnessed firsthand the frustrations faced by researchers, peers, and faculty in navigating the ever-expanding and fragmented landscape of scholarly literature. The challenge of locating specific research, discovering related work, and avoiding redundant studies is a common hurdle that slows down the pace of learning and innovation.

We specifically chose this problem of knowledge discovery in research institutions because it sits at the perfect intersection of our academic interests and practical impact. It allows us to move beyond theoretical concepts and build a functional system that leverages practical AI techniques—like NLP and machine learning—to create immediate value. The opportunity to develop a tool that can save researchers significant time, foster new collaborations by connecting related work, and ultimately accelerate scientific progress in critical areas like public health is what truly drives us. This project represents a meaningful culmination of our studies, enabling us to translate our knowledge of databases, web technologies, and AI into a solution that addresses a genuine need within the Ethiopian research ecosystem and beyond.

Our motivation extends beyond creating another digital repository. We aim to solve Ethiopia's specific "research redirection problem" where accessing local research requires navigating through international paywalls and broken links. By building a centralized, AI-enhanced platform that hosts research directly, we create:

- Equitable access for all Ethiopian researchers and students
- National research visibility through unified analytics

- Knowledge sovereignty by hosting content locally

This represents a paradigm shift from passive linking to active hosting, and from global generic platforms to locally-tailored intelligence.

1.4. Statement of the Problem

Research institutions in Ethiopia currently face a dual challenge in knowledge management and dissemination. First, there is an inefficient management of the escalating volume of scholarly publications and datasets, where existing systems lack the intelligence to understand contextual relationships within research data. Second, and more critically, there exists a structural fragmentation in how research is accessed. Institutional websites across Ethiopia primarily serve as mere directories that redirect users to external publisher portals (Elsevier, Springer, Wiley, etc.), creating multiple barriers:

1. Access Barriers: Many redirected links lead to paywalled content inaccessible without expensive institutional subscriptions.
2. Link Instability: External publisher URLs frequently change or become inactive, resulting in broken links.
3. Fragmented Discovery: Researchers cannot perform unified searches across multiple Ethiopian institutions.
4. No National Research Identity: There is no consolidated view of Ethiopia's research output, making it impossible to analyze national research trends.
5. Data Sovereignty Concerns: Ethiopian research remains hosted on international servers with limited local control.

This fragmented redirection model leads to inefficient search processes, duplicated research efforts, missed collaborative opportunities, and ultimately slows the pace of scientific progress within Ethiopia. There is an urgent need for a centralized, locally-hosted platform that provides direct access to Ethiopian research while incorporating intelligent discovery tools.

1.5. Objective of the Study

1.5.1. General Objective

To design and develop a knowledge management system that enhances discovery of health research papers through automated organization, medical-domain search optimization, and intelligent recommendations.

1.5.2. Specific Objective

1. To develop a web-based application with user authentication for secure access.
2. Develop a document upload system supporting PDFs with automated metadata extraction.
3. To deploy a search engine (Elasticsearch) enabling keyword and filter-based discovery across all uploaded documents.
4. To develop institution and author profile pages that aggregate their publications within the system.
5. To build a researcher dashboard where users can save searches and receive alerts for new matching documents.
6. To integrate NLP-based tagging (TF-IDF + LDA) for auto-categorization of documents.
7. To provide a content-based recommendation system suggesting related papers.
8. To implement Elasticsearch for powerful and efficient semantic search capabilities.

1.6. Feasibility Analysis

1.6.1. Operational Feasibility

The proposed system is designed to be highly user-friendly, requiring minimal training for researchers to use. It directly addresses the operational inefficiencies of current knowledge management practices by automating manual tasks like tagging and complex searching. The intuitive interface and logical workflow will ensure it integrates smoothly into the daily routines of a research institution, enhancing productivity rather than disrupting it.

1.6.2. Technical Feasibility

The project is technically feasible. Our team possesses the necessary skills in Python, web development, and the fundamentals of machine learning. The proposed technology stack—including Python, scikit-learn, Elasticsearch, and modern NLP libraries (spaCy, NLTK)—is well-documented, widely supported, and appropriate for our implementation approach. The use of established techniques like TF-IDF and LDA, rather than complex deep learning models, ensures the machine learning components are achievable within our timeline and computational resources.

1.6.3. Economic Feasibility

The development of this system is highly cost-effective, primarily relying on open-source software and existing hardware. The main anticipated costs are for deployment and professional presentation, as detailed below:

Resource	Description	Estimated Cost
Web Hosting	A cloud server to deploy the backend application, database, and search engine for 24/7 access.	\$100 - \$150 / year
Domain Name	A professional domain name for user-friendly access.	~\$15 / year
Reference Materials	Budget for potential online tutorials or specialized resources.	~\$50 (One-time)
Total		~\$165 - \$215

All core development software (Python, Flask, React, Elasticsearch) and hardware (personal laptops) are open-source and already available, resulting in no additional cost

1.6.3.1. Cost-Benefit Analysis

The economic value of this project significantly outweighs its minimal cost. The primary benefit is the substantial time saved for researchers and academics, who currently spend hours manually searching for literature. By streamlining knowledge discovery, the system allows these highly skilled individuals to reallocate valuable time toward core research activities, analysis, and collaboration. This increase in research efficiency and

potential acceleration in scientific output represents a return on investment that far exceeds the modest development and hosting costs.

1.6.4. Schedule Feasibility

The project is meticulously planned to be completed within the two-semester timeframe of the Final Year Project. A detailed schedule with specific milestones—including the submission of the Requirements Specification, core development completion, AI integration, and final delivery—has been created to ensure the timely and sequential completion of all objectives, with built-in buffers to accommodate any unforeseen challenges.

1.7. Scope of the Project

This project focuses on the design, development, and implementation of a prototype intelligent knowledge management system tailored for a research institution environment.

1.7.1. In-Scope

1. User authentication (registration and login) with role-based access control (Researcher, Institutional Admin, System Admin).
2. Direct document hosting and management for various research document formats.
3. Cross-institutional search and aggregation capabilities.
4. Automated tagging and categorization using TF-IDF and LDA topic modeling trained on Ethiopian research.
5. Enhanced semantic search functionality with medical synonym support and Ethiopian context awareness.
6. Content-based recommendation system using cosine similarity.
7. PDF text extraction and automatic metadata generation.
8. Management of institution and author profiles with national research analytics.
9. Personalized user dashboards for managing saved searches and preferences.
10. Advanced search filtering options based on institution, author, publication date, and topic.

1.7.2. Out-of-Scope

1. Real-time collaboration features (e.g., live document editing or commenting).
2. Integration with external citation management tools (e.g., Zotero, Mendeley).
3. Full-scale deployment across all Ethiopian universities nationwide.
4. Implementation of a comprehensive knowledge graph.
5. Advanced document summarization features.
6. Complex neural network-based models for processing.
7. Processing of scanned image-based PDF documents (non-selectable text).
8. Mobile application development (web-only platform).

1.7.3. System Uniqueness and Innovation

What distinguishes our system from existing solutions:

1. **Direct Hosting Model:** Unlike current institutional sites that redirect to publishers, our system hosts PDFs directly (with appropriate permissions), ensuring permanent access without broken links.
2. **Cross-Institutional Aggregation:** We unify research from multiple Ethiopian institutions into a single searchable platform while preserving institutional identity.
3. **Ethiopian Research Intelligence:** Our AI models are specifically trained on Ethiopian research corpora, understanding local terminology and context better than generic international systems.
4. **National Research Mapping:** We provide unique analytics on Ethiopia's research landscape—currently unavailable through any existing platform.
5. **Hybrid Publishing Model:** Supports both institutional submissions (curated) and individual researcher uploads, creating a comprehensive national database.

1.8. Significance of the Project

This project holds substantial importance for the research ecosystem by addressing fundamental challenges in knowledge management and discovery. Its primary significance lies in transforming how research information is accessed and utilized, moving from a passive storage model to an active discovery platform.

The system will deliver significant value through several key impacts:

1. **Enhanced Research Efficiency:** By providing intelligent search capabilities and automated organization, the system will dramatically reduce the time researchers spend looking for relevant literature. This time savings can be redirected toward actual research activities, potentially accelerating the pace of scientific discovery.
2. **Mapping the Research Landscape:** A key innovation is the organization of research by institution and author. This will effectively map Ethiopia's health research landscape, making visible the contributions of different organizations and individual researchers. Such mapping is crucial for identifying research trends, gaps, and opportunities for collaboration.
3. **Solving the Redirection Problem:** By hosting research directly rather than linking to external sites, we eliminate access barriers, broken links, and dependency on international platforms, ensuring permanent access to Ethiopian research.
4. **Fostering Collaboration:** By highlighting key contributors and their work across institutions, the system naturally fosters interdisciplinary and inter-institutional collaboration. Researchers can easily identify others working on similar problems, leading to potential partnerships that might not have been discovered through conventional means.
5. **Knowledge Preservation and Accessibility:** The system serves as a centralized, intelligent archive that preserves institutional knowledge and makes it readily accessible to new generations of researchers, ensuring valuable insights are not lost over time.

6. **Supporting Evidence-Based Decision Making:** For policymakers and public health officials, the system provides efficient access to consolidated research findings, enabling more informed decision-making based on comprehensive evidence.

1.9. Target Beneficiaries

The system is designed to serve a diverse range of users within the academic and research ecosystem, with varying levels of engagement and benefit:

Primary Beneficiaries:

- **Researchers & Scientists:** Professionals actively conducting research who need efficient access to relevant literature and related works.
- **PhD & MSc Students:** Graduate students requiring comprehensive literature review capabilities for their theses and dissertations.
- **Faculty Members:** Academic staff needing to stay current with research developments in their field and identify potential collaborators.

Secondary Beneficiaries:

- **Librarians & Information Specialists:** Institutional staff responsible for managing research repositories and assisting users in finding information.
- **Academic Administrators:** Department heads and research directors who need to monitor institutional research output and identify research trends.
- **Research Institutions:** Organizations seeking to showcase their research portfolio and enhance the visibility of their academic contributions.

Indirect Beneficiaries:

- **Policy Makers:** Government and public health officials who need evidence-based research to inform policy decisions.
- **The General Public:** Ultimately, society benefits from accelerated research progress and more efficient knowledge translation into practical solutions.

1.10. Methodology

The project will follow a structured methodology integrating software development life-cycle with machine learning pipeline development, specifically tailored for an AI-based knowledge management system.

1.10.1. Fact-Finding Techniques

We will employ multiple fact-finding approaches including analysis of existing research repository systems, informal interviews with students and faculty members to understand user needs and challenges, and practical observation of current research discovery work-flows.

1.10.2. Data Source and Preprocessing

The machine learning models will be trained on publicly available datasets of academic papers, primarily the CORD-19 dataset and arXiv collections, which provide substantial corpora of research text data suitable for NLP model training and system validation.

1.10.3. Data Type

The system will primarily process unstructured text data extracted from research documents in PDF format, along with structured metadata including author information, institutional affiliations, publication dates, and document classifications.

1.10.4. Data Cleaning

This crucial phase involves removing incorrect, corrupted, or duplicate data within datasets. Specific steps will include special character removal, text normalization, handling missing values, and standardization of document formats to ensure data quality for model training.

1.10.5. Data Labeling or Annotation

For topic modeling implementation, data labeling will be performed automatically using the Latent Dirichlet Allocation (LDA) algorithm to identify key topics within research

documents and assign meaningful categorical labels that enable intelligent content organization.

1.10.6. Systems Analysis and Design

We will employ UML modeling techniques including Use Case diagrams, Activity diagrams, and Sequence diagrams to comprehensively capture system requirements and architect the solution, ensuring all functional and non-functional requirements are properly addressed.

1.10.7. Development Tools

Category	Software/Hardware Tools
Platform	Windows / Linux
AI/ML Tools	Python, scikit-learn, NLTK, spaCy, Jupyter Notebook
Backend	Django or Flask (Python web frameworks)
Database	PostgreSQL or MySQL
Search Engine	Elasticsearch
Frontend	HTML, CSS, JavaScript (framework like React or Vue.js)
Editors	VS Code, PyCharm
Hardware	Standard Laptops/PCs, potentially a GPU for model training

1.10.8. Training and Evaluation

1.10.8.1. Training/Testing Data Split

The dataset will be strategically partitioned using an 80/20 split, with 80% allocated for model training and 20% reserved for validation and testing to ensure model robustness and prevent overfitting.

1.10.8.2. Performance/Evaluation

Model effectiveness will be quantitatively assessed using standard metrics including precision, recall, and F1-score for classification tasks, and cosine similarity scores for evaluating the recommendation engine's performance.

1.10.9. Implementation Strategy

A direct implementation approach is planned, where the fully developed system will be deployed in a controlled environment for comprehensive user acceptance testing and validation against specified requirements.

1.11. Limitation of the Project

The project development acknowledges several inherent limitations that define the scope and operational boundaries of the system:

Scope of Data: The system is designed to process English-language PDF documents containing extractable text. Scanned PDFs (image-based documents) will not be processed correctly due to the absence of optical character recognition (OCR) capabilities. The system primarily focuses on textual content and does not handle images, datasets, or complex document formatting elements.

Machine Learning Simplicity: Our machine learning approach utilizes established, interpretable techniques (TF-IDF, LDA) rather than complex deep learning models. This makes the system more transparent and suitable for our available dataset size and computational resources, though it may be less sophisticated than commercial, state-of-the-art systems.

Extraction Limitations: The accuracy of automatic title and abstract extraction is dependent on the consistency of document formatting in uploaded research papers. Inconsistently formatted documents may require manual user verification and correction of the extracted metadata.

Proof-of-Concept Nature: This project is developed as a functional prototype to demonstrate the core concept's viability. As such, it will not incorporate the extensive security, maintenance, error-handling, and scalability features expected of a commercial-grade, production-ready enterprise application.

1.12. Team Composition

S.No	Full Name	ID.No.	Email	Responsibility
1.	Soreti Tesfaye	RU1932/15	soretitesfaye16@gmail.com	Team Leader, Backend, Elasticsearch, Search Optimization
2.	Kibrenesh Tilahun	RU1237/15	Kebrontilahun1@gmail.com	ML, TF-IDF & LDA dev
3.	Yordanos Solomon	RU2219/15	yordisolomon123@gmail.com	Frontend Development, UI/UX Design
4.	Hermela Tefera	RU1066/15	Hermelateffera186@gmail.com	Database Management
5.	Worknesh Chalchisa	RU2121/15	Worke2119@gmail.com	System Testing, Documentation

Chapter Two: Requirement Analysis and Specification

2.1. Introduction of Existing System

The current process for Ethiopian health researchers to find and access relevant research documents is largely decentralized and manual. The primary platform for discovery is often the Ethiopian Public Health Institute (EPHI) website, which hosts publications in PDF format, typically within a simple directory structure or a basic list. Researchers must rely on manually browsing these lists or using generic website search functions that lack advanced filtering capabilities. Other sources include the institutional repositories of universities, which are often isolated and not interconnected, creating knowledge silos.

An analysis of the limitations of this existing system using the PIECES framework reveals the following critical problems

- **Performance:** The system's performance is inadequate for scholarly research. Researchers spend a significant amount of time manually browsing through long, unsorted lists of documents to find relevant work. This lack of a specialized search engine leads to inefficient and time-consuming information retrieval processes.
- **Information and Data:** The quality and structure of the information are poor. The output is often limited to the document itself, with no consolidated view of an author's entire body of work or an institution's research output. Furthermore, there is no structured data input for authors or institutions, leading to inconsistencies such as varied spellings of an author's name across documents. This results in data redundancy, where the same document might be uploaded to multiple repositories without a central index.
- **Economics:** The manual effort required from researchers represents a significant hidden cost in terms of time and productivity. The inefficiency of the discovery process directly impacts the pace of research and innovation.

- **Control and Security:** The existing systems often lack robust administrative controls. There is no centralized mechanism for content management, user role definition, or ensuring data consistency across the national research landscape.
- **Efficiency:** The process is highly inefficient. Researchers cannot save their search queries to be notified of new, relevant publications. They must repeatedly perform the same manual searches to stay updated in their field.
- **Services:** The existing system provides a basic document hosting service but fails to offer the value-added services that are standard in modern digital libraries. It lacks features such as personalized dashboards, author profiling, recommendation engines, or institutional analytics that actively aid in knowledge discovery.

2.2. Proposed System

Our proposed system is an AI-enhanced, centralized research repository and digital library specifically designed for Ethiopian health research. It will provide a sophisticated, user-centric platform for discovering, managing, and analyzing research documents, directly addressing the limitations identified in the PIECES analysis.

A fundamental principle of our system is fully open access with no barriers. Unlike existing systems that may require registration or institutional subscriptions, our platform will allow any user to immediately download any research document without any registration or login requirements. This commitment to true open access ensures that Ethiopian research reaches the widest possible audience, including students, policymakers, rural researchers, and the general public who may not have institutional affiliations or email addresses.

The system will ingest documents from various sources, and by leveraging AI and natural language processing, it will automatically extract and index key metadata, including title, abstract, publication date, author names, and their affiliated institutions. This automated process ensures that the repository's data is structured, consistent, and ready for intelligent analysis.

The key features of the proposed system include:

- ◆ **Multi-Institutional Centralization:** A unified platform aggregating research from multiple Ethiopian institutions while maintaining institutional identity through dedicated profile pages.
- ◆ **Open Access:** Immediate PDF downloads for all users without any registration requirements
- ◆ **Advanced Search and Filtering:** Users can perform powerful semantic searches and then refine results using multiple filters, including publication date range, author name, institution, and topic.
- ◆ **Multi-Tier User Management:** A comprehensive role-based system with different access levels for researchers, institutional administrators, system administrators, and public users.
- ◆ **Author and Institution Profiling:** Each author's name on a document will be a clickable link leading to a dedicated Author Profile Page. Similarly, institutions will have profile pages showcasing their collective research output.
- ◆ **Document Submission Workflow:** A structured submission process with approval workflows for institutional verification and quality control.
- ◆ **User Dashboard and Personalization:** Registered users will have a personal dashboard with saved searches, alerts for new matching documents, and personalized recommendations.
- ◆ **Ethiopian Context Intelligence:** AI models specifically trained on Ethiopian research corpora for better contextual understanding and relevance.

2.3. Requirements

This section formally lists the functional and non-functional requirements of the proposed system. These requirements are derived from the project objectives and define the criteria for a successful implementation.

2.3.1. Functional Requirements

Functional requirements specify what the system shall do.

1. Open Access and Public Usage

- The system shall allow any user (authenticated or anonymous) to browse and search all documents
- The system shall allow any user to download full PDF documents immediately without any registration or authentication requirements
- The system shall track download statistics anonymously without requiring user identification
- The system shall provide anonymous usage analytics showing popular documents and search trends

2. User Management and Authentication (for Contributors)

- The system shall allow new users to register with different roles (Researcher, Institutional Admin, System Admin).
- The system shall require authentication only for content contribution and personalized features, not for access or downloads
- The system shall implement comprehensive role-based access control distinguishing between Public User, Researcher, Institutional Admin, and System Admin.

3. Document Management

- The system shall allow authenticated Researchers to upload new research documents with institutional affiliation.
- The system shall require Moderator approval for all uploaded documents before public visibility
- The system shall provide Moderators with tools to review documents for quality, appropriateness, and proper categorization
- The system shall allow Moderators to approve, request revisions, or reject uploaded documents

- The system shall notify uploaders of moderation decisions with appropriate feedback

4. Search and Discovery

- The system shall provide semantic search capabilities across all hosted documents.
- The system shall allow users to filter search results by publication date, author, institution, and topic.
- The system shall provide content-based recommendations for related documents using cosine similarity.
- The system shall display download statistics alongside search results to indicate document popularity

5. Institutional and Author Management

- The system shall allow Institutional Admins to create and manage their institution's profile.
- The system shall automatically generate author profile pages listing all their publications in the system.
- The system shall provide institutional analytics showing research output trends and download statistics
- The system shall allow Institutional Admins to verify documents uploaded by their institution's members

6. Quality Assurance and Moderation

- The system shall implement a moderation queue for newly uploaded documents
- The system shall provide Moderators with tools to check document quality, formatting, and appropriatenes
- The system shall allow Moderators to manage system-wide taxonomies and controlled vocabularies
- The system shall maintain an audit trail of all moderation decision

7. Personalization Features

- The system shall provide registered users with a personalized dashboard.
- The system shall allow users to save search queries with custom names
- The system shall notify registered users via their dashboard when new documents match their saved searches
- The system shall provide document recommendation feeds based on user activity and interests.

2.3.2. Non-Functional Requirements

Non-functional requirements specify how the system should perform.

1. Performance

- The system shall load search results and primary web pages in under 3 seconds under normal load conditions.

2. Usability

- The user interface shall be intuitive and easy to navigate for users with basic computer literacy, requiring minimal training.
- The system shall provide contextual help and tool-tips for complex features.

3. Security

- The system shall encrypt user passwords using industry-standard hashing algorithms.
- The system shall implement session management with automatic timeout.
- Role-based access shall be strictly enforced at all access points.

4. Availability

- The system shall be available 24/7, with acceptable, pre-scheduled downtime for maintenance.

5. Scalability

- The system architecture shall be designed to accommodate a growing number of documents and users without significant performance degradation.

6. Open Access Compliance

- The system shall clearly communicate the open access policy on all pages
- No barriers (registration, login, payment) shall be required for accessing or downloading content
- Usage statistics shall be publicly available to demonstrate the platform's impact

2.4. System Models (SRS)

This section provides a detailed model of the proposed system using standard Unified Modeling Language (UML) diagrams and user interface prototypes. The purpose of these models is to provide a clear and unambiguous specification of the system's requirements, structure, and behavior, serving as a blueprint for the design and implementation phases.

2.4.1. Use Case Diagram

The Use Case Diagram provides a high-level overview of the interactions between the system and its external actors. It establishes the functional boundaries of the system by illustrating the core capabilities and which actor is responsible for initiating each function

Actors:

1. Public User/Guest: Any user accessing the system without authentication who can:

- Browse all documents and metadata
- Search across the entire repository
- Download any PDF document immediately without registration

- View author and institution profiles
- Use all filtering and sorting options

2. Registered User/Researcher: Authenticated user who can:

- Everything Public Users can do
- Upload research documents for inclusion in the repository
- Save personalized searches and receive alerts
- Claim authorship of existing documents
- Track their upload history and status
- Receive personalized recommendations

3. Institutional Administrator: Authenticated user representing an Ethiopian research institution who can:

- Everything Registered Users can do
- Manage their institution's public profile page
- Verify documents uploaded by members of their institution
- View detailed analytics about their institution's research impact
- Manage member affiliations and institutional information

4. Moderator/Content Manager: System staff responsible for quality assurance who can:

- Review all newly uploaded documents for quality and appropriateness
- Approve, reject, or request revisions on submissions
- Manage system taxonomies and categorization schemes
- Ensure compliance with content guidelines

- Handle copyright and licensing verification

5. System Administrator: Technical staff with full system control who can:

- Everything Moderators can do
- Manage all user accounts and permissions
- Configure system settings and parameters
- Perform system maintenance and backups
- Monitor system performance and usage statistics

The primary use cases for the system are:

For Public Users: Browse Documents, Search Repository, Download PDFs, View Document Details, View Author Profiles, View Institution Profiles

For Registered Users: All Public User functions plus: Register/Login, Upload Document, Save Search Query, View Personalized Dashboard, Receive Search Alerts, Claim Authorship

For Institutional Admins: All Registered User functions plus: Manage Institution Profile, Verify Institution Documents, View Institution Analytics, Manage Member Affiliations

For Moderators: Review Document Submissions, Manage Taxonomies, Ensure Content Quality, Handle Copyright Compliance, Manage Content Guidelines

For System Admins: All Moderator functions plus: Manage Users, System Configuration, Backup Management, Performance Monitoring

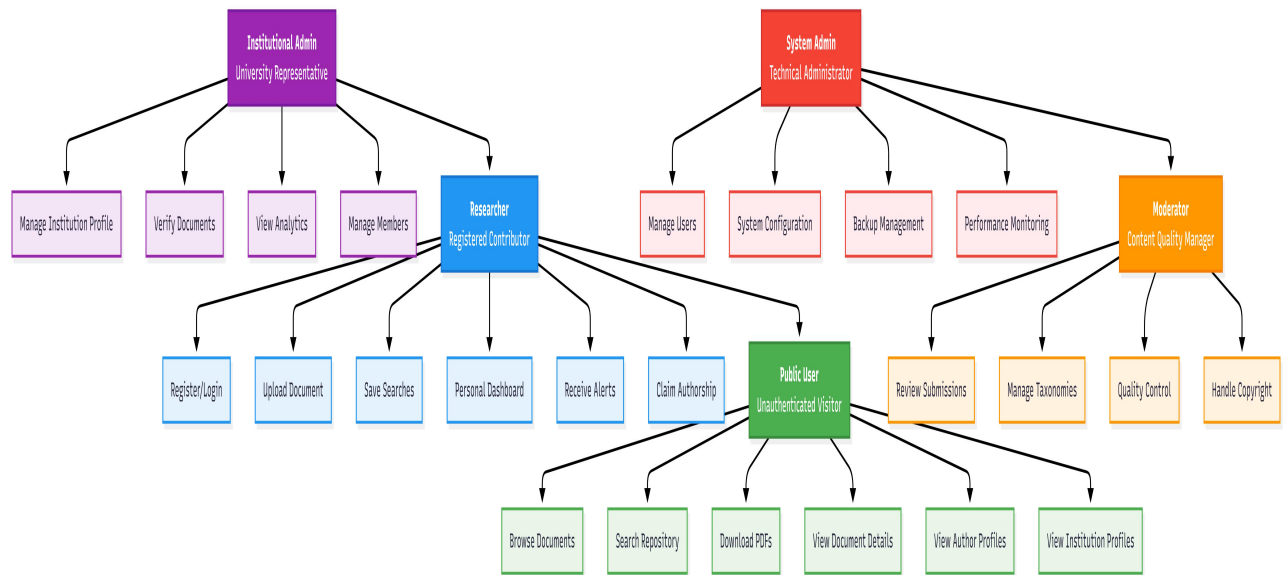


Figure 2.1: Use Case Diagram for the Intelligent Knowledge Management System

2.4.2. Use Case Documentation

Use Case Documentation provides a detailed, step-by-step description of the interaction between an actor and the system for a specific use case. This clarifies the exact behavior and flow of the system from the user's perspective. Below are documented examples for the key use cases of the system

Use Case Name	Instant PDF Download Without Registration
Actor(s)	Public User/Guest
Description	Allows any user to immediately download any research document without any authentication requirements.
Pre-conditions	User has internet access and has located a document of interest
Main Flow	1. User finds document through search, browsing, or direct link. 2. User clicks the "Download PDF" button. 3. System immediately serves the PDF file to the browser. 4. System increments anonymous download counter for the document. 5. User saves the file locally or opens it directly.
Alternative Flow	3a. If file is temporarily unavailable, system displays friendly error

	message with expected restoration time.
Post-conditions	User has the complete research paper; download statistics are updated.

Table 2.1: Use Case for Anonymous PDF Download (Core Open Access Feature)

Use Case Name	Document Upload and Moderation Process
Actor(s)	Researcher, Moderator
Description	Allows researchers to submit documents that undergo quality review before becoming publicly available.
Pre-conditions	User is registered and logged in as Researcher
Main Flow	1. Researcher navigates to upload page. 2. Researcher fills required information (PDF, title, abstract, authors, optional institution). 3. System validates and processes document (extracts text, generates metadata). 4. Document enters "Pending Moderation" queue. 5. Moderator reviews document for: academic quality, appropriateness, formatting, metadata accuracy. 6. Moderator takes action: a. Approve: Document becomes publicly available for immediate download by anyone. b. Request Revisions: Returns to researcher with specific feedback. c. Reject: Removes from system with explanation. 7. Researcher receives notification of decision.
Post-conditions	Document is either publicly available or returned for revision.

Table 2.2: Document Submission with Moderation Workflow

Use Case Name	Institutional Document Verification
Actor(s)	Institutional Administrator
Description	Allows institutional administrators to verify documents uploaded by members of their institution.
Pre-conditions	User is logged in as Institutional Admin, Document claims affiliation with their institution
Main Flow	1. System notifies Institutional Admin of new document claiming their affiliation. 2. Admin reviews document details and author information. 3. Admin verifies if author is indeed affiliated with their institution. 4. Admin takes action: a. Verify: Confirms institutional affiliation. b. Reject Affiliation: Removes institution claim from document. 5. Document proceeds to general moderation queue. 6. Verification status is displayed on document page.
Post-conditions	Document's institutional affiliation is either verified or removed.

Table 2.3:Institutional Document Verification

Use Case Name	Systematic Content Moderation
Actor(s)	Moderator/Content Manager
Description	Ensures all content meets quality standards before becoming publicly available in the open repository.
Pre-conditions	User has Moderator role, Documents are in moderation queue
Main Flow	1. Moderator accesses moderation dashboard. 2. Views queue of pending documents

	sorted by submission date. 3. For each document: a. Reviews PDF quality (readability, completeness). b. Checks metadata accuracy. c. Verifies academic appropriateness (not spam, commercial content, etc.). d. Ensures proper categorization and tagging. e. Checks for copyright compliance. 4. Takes moderation action with comments. 5. System updates document status and notifies uploader.
Post-conditions	Repository maintains high-quality, appropriate academic content.

Table 2.4: Content Moderation and Quality Control

Use Case Name:	Save Search Query
Actor(s):	Researcher
Description:	Allows a researcher to save the parameters of their current search (keywords and filters) for future use and to receive automated alerts.
Main Flow:	1. The researcher performs a search with specific keywords and filters. 2. The researcher clicks the “Save Search” button on the results page. 3. The system prompts the user to provide a name for the search. 4. The user provides a name (e.g., “Malaria Research 2020+”) and confirms. 5. The system validates the input and saves the search query parameters to the user’s profile in the database. 6. The system displays a confirmation message to the user.

Post-conditions:	The search query is stored and is visible on the user's dashboard.
------------------	--

Table 2.5: Use Case for Saving a Search Query

Use Case Name:	Upload Document
Actor(s):	Researcher
Description:	Allows an authenticated researcher to upload a new PDF document, which the system will then automatically process and index.
Pre-conditions:	The user must be logged into the system.
Main Flow:	1. The user navigates to the "Upload Document" page. 2. The system displays a form with fields for file selection and associating an Institution. 3. The user selects a PDF file from their local machine. 4. The user selects the affiliated Institution from a pre-populated list. 5. The user clicks the "Submit" button. 6. The system validates the file and begins the AI processing pipeline (as shown in the Activity Diagram). 7. Upon successful processing, the system displays a "Document successfully uploaded" message.
Alternative Flow:	6a. If the selected file is not a valid PDF, the system displays an error message and the use case ends.
Post-conditions:	The document is processed, indexed, and is now discoverable via the search functionality.

Table 2.6: Use Case for Uploading a Document

Use Case Name:	Manage Institutions
Actor(s):	Administrator
Description:	Allows an administrator to add, edit, and delete the institution profiles that are available in the system.
Pre-conditions:	The user must be logged in with the Administrator role.
Main Flow (Add):	1. The administrator navigates to the “Manage Institutions” dashboard. 2. The admin clicks the “Add New Institution” button. 3. The system displays a form with fields for Institution Name and Description. 4. The admin fills out the form and clicks “Save.” 5. The system validates the data and adds the new institution to the database.
Main Flow (Edit):	1. From the institutions list, the admin clicks the “Edit” button next to an institution. 2. The system displays the form pre-filled with the existing data. 3. The admin makes the necessary changes and clicks “Save.” 4. The system updates the record in the database.
Post-conditions:	The list of institutions in the system is updated.

Table 2.7: Use Case for Managing Institutions

2.4.3. Activity Diagram

The Activity Diagram illustrates the workflow and the sequence of actions between the user and the system for a specific process. The diagram below models the workflow for the "Document Upload and AI Processing" scenario, which is a core automated function of the system.

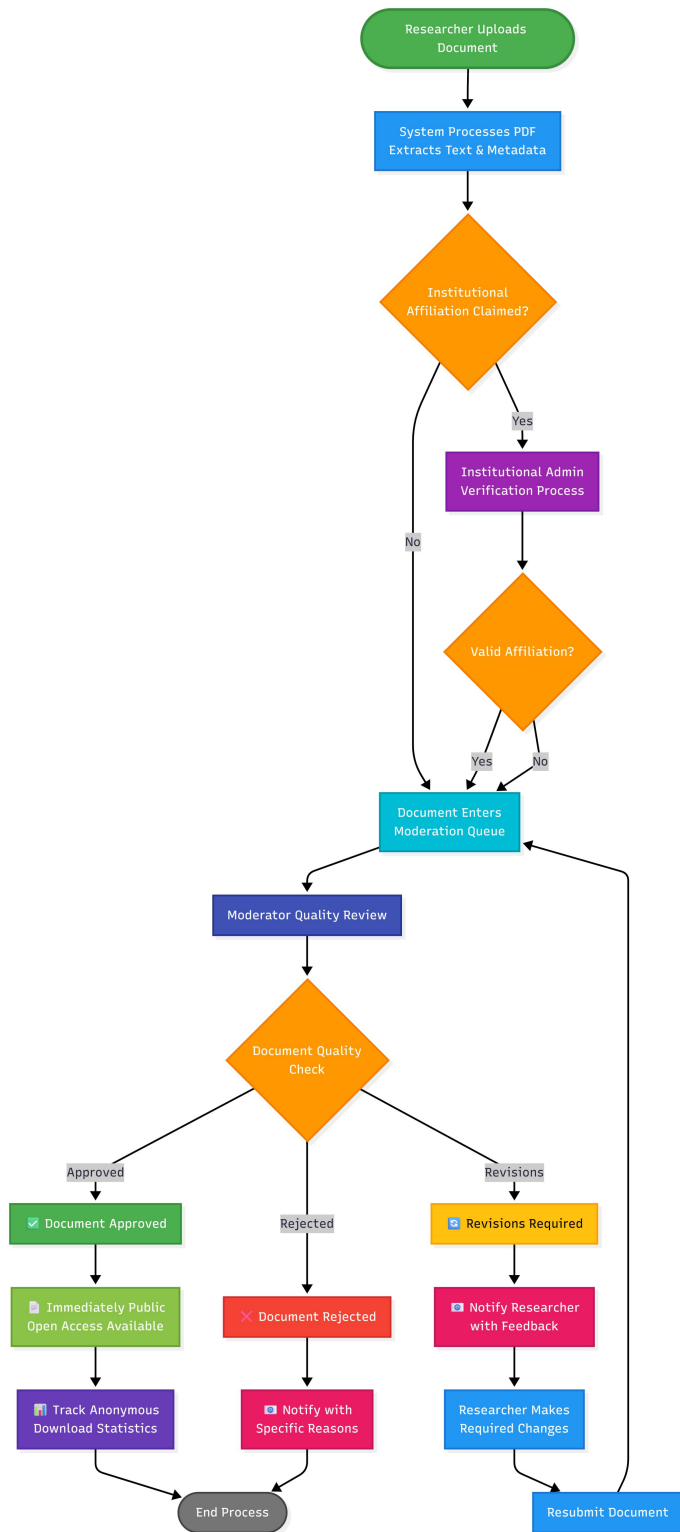


Figure 2.2: Activity Diagram for Document Upload, Processing, and Moderation Workflow

2.4.4. Sequence Diagram

The Sequence Diagram shows the time-ordered interaction between system components during a specific operation. The diagram below details the sequence of interactions when a user filters search results by a specific institution, clarifying how backend components collaborate to fulfill the request.

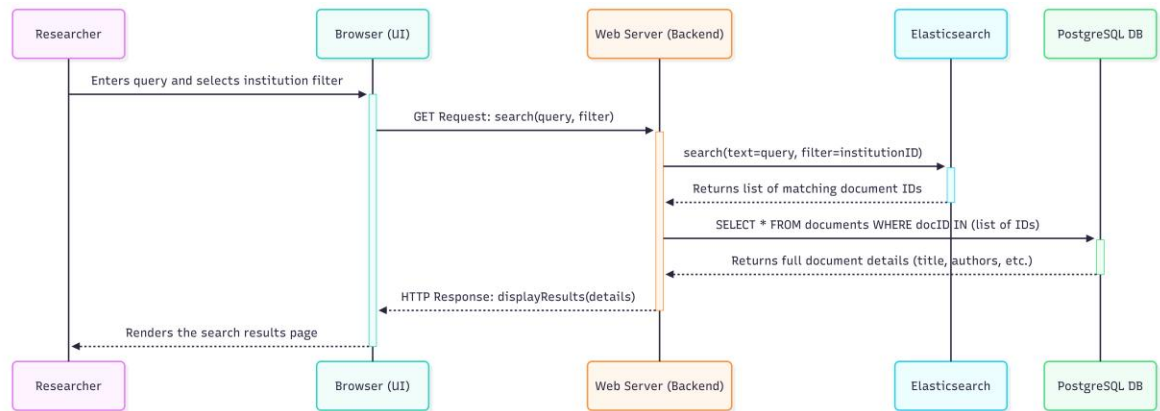


Figure 2.3: Sequence Diagram for a Filtered Search Request

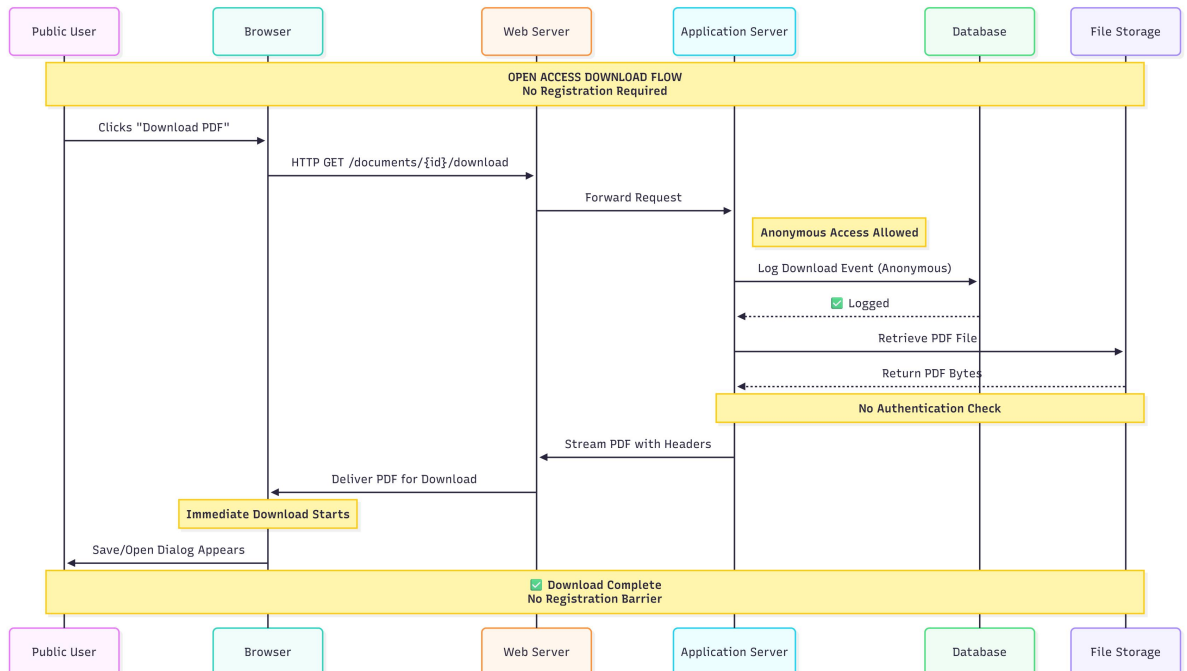


Figure 2.4: Sequence Diagram for Instant PDF Download Process (Open Access Model)

2.4.5. Class Diagram

The Class Diagram represents the static structure of the system, detailing the main classes, their attributes, and the relationships between them.

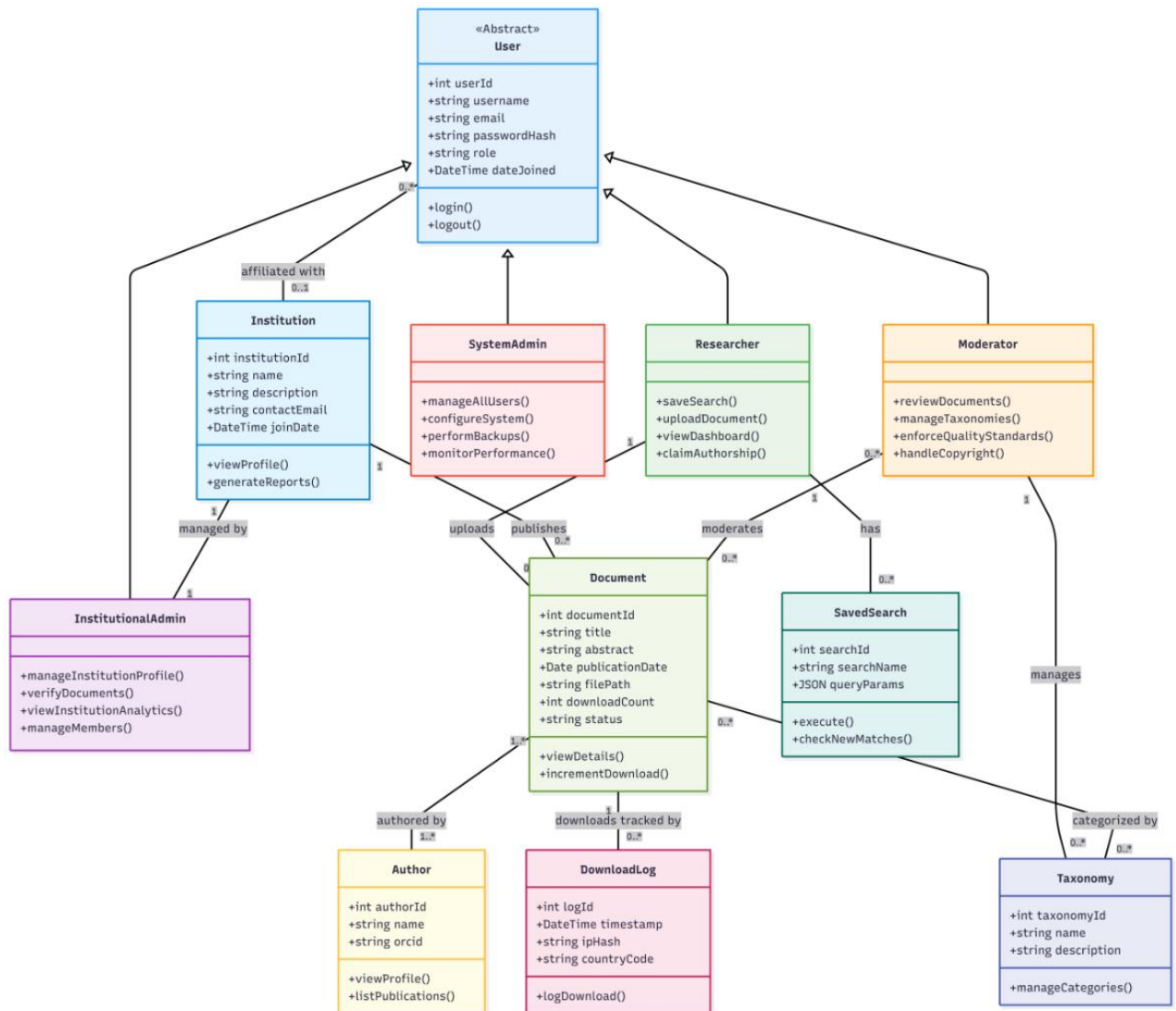


Figure 2.5: System Class Diagram

2.4.6. UI Prototyping

User Interface (UI) prototypes provide a visual guide for the look and feel of the system. They help to translate the functional requirements into a tangible design. Below are mockups of the key pages of the AI-Enhanced Research Repository.

AI-Enhanced Research Repository

Author

Institution

Publication Date

Figure 2.5: UI Mockup of the Main Search and Filter Page

Filters

Author

Institution

Search Results (12 found)

Impact of Malaria Interventions in Ethiopia

This paper reviews the effects of recent malaria control strategies...

Authors: A. Kebede, L. Demeke | Institution: Jimma University | Date: 2023

Maternal Health Trends in Addis Ababa

A comprehensive study on the changing trends of maternal healthcare...

Authors: S. Bekele | Institution: EPHI | Date: 2022

Figure 2.6: UI Mockup of the Search Results Page

My Dashboard

Welcome back, Soret Tesfaye

Saved Searches

"
Malaria Research 2020+
"
3 new
Run

"
Jimma University Health
"
1 new
Run

Recent Document Feed

Dengue Outbreak Patterns in the Eastern Region

Authors: T. Alemayehu, F. Girma | Institution: EPHI

Matches your saved search: "Malaria Research 2020+"

Cardiovascular Health Survey in Urban Ethiopia

Authors: B. Tadesse | Institution: Jimma University

Matches your saved search: "Jimma University Health"

Figure 2.7: UI Mockup of the Researcher Dashboard

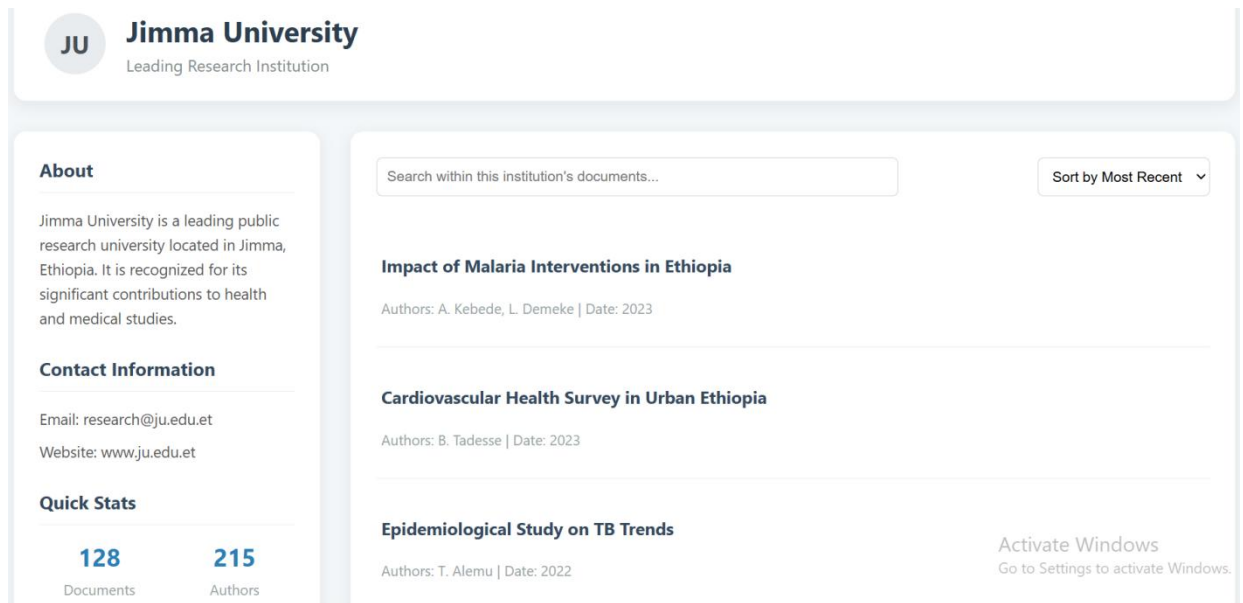


Figure 2.8: UI Mockup of an Institution Profile Page

Chapter Three: System Design

3.1. Introduction

This chapter outlines the high-level design of the Intelligent Knowledge Management System. It serves as the bridge between the requirement analysis phase (Chapter 2) and the future implementation phase. The requirements gathered and the system models specified in the previous chapter are used as the primary input to create a concrete architectural and technical blueprint. This blueprint will guide the development team in constructing a system that is robust, scalable, and fully aligned with the project's objectives, particularly emphasizing the open access model and multi-actor architecture.

3.2. Purpose of System Design

The purpose of the system design phase is to produce a detailed and comprehensive plan for building the software. This involves defining the system's architecture, components, modules, interfaces, and data structures. A well-defined design ensures that all components will integrate seamlessly, meet the specified functional and non-functional requirements, and provide a solid foundation for future maintenance and expansion. This design document aims to minimize ambiguity and provide a clear roadmap for the development process, with particular attention to the unique requirements of an open access, multi-institutional research repository.

3.3. Design Goals

The design of the system is driven by several key goals, which are derived directly from the non-functional requirements identified in Chapter 2. These principles will guide all architectural and implementation decisions:

Open Access First: The architecture must prioritize frictionless access to research content. All users, regardless of authentication status, should be able to discover and download research documents with minimal barriers, while maintaining appropriate quality controls through backend moderation workflows.

Multi-Tenant Scalability: The system must support multiple institutions as distinct tenants within a unified platform. Each institution should have its own identity and administrative controls while sharing common search and discovery infrastructure.

Maintainability: The system will be built using a modular approach, with a clear separation of concerns. This ensures that the codebase is easy to understand, debug, and update in the future.

Performance: The system must be responsive and efficient, particularly in its core search and recommendation functionalities. The design will prioritize fast data retrieval and efficient processing pipelines to meet the performance targets defined in the requirements.

Security: While providing open access, the system must ensure proper access controls for content submission and moderation, protecting intellectual property and maintaining data integrity across institutional boundaries.

Usability: While primarily a concern of the user interface, the backend design must support a smooth and intuitive user experience. This includes providing fast API responses and consistent data handling.

3.4. Deployment Diagram

The deployment diagram illustrates the physical layout of the system's hardware and the distribution of its software components. Our system will be deployed using a scalable cloud infrastructure to ensure high availability and reliability, with particular emphasis on handling the anticipated load from open access downloads.

The deployment consists of three primary environments:

Client Environment: Represents the end-user's computing devices including desktops, laptops, and mobile devices running standard web browsers. No specialized software is required beyond a modern web browser with JavaScript support..

Application Server Environment: Hosted on a cloud virtual private server (VPS) with the following components:

- Web Server (Nginx): Handles HTTP/HTTPS requests, serves static files, and acts as reverse proxy
- Application Server (Gunicorn/Flask): Python-based application server running the core business logic
- Message Queue (Redis): Manages background tasks including document processing and notification delivery
- Background Workers: Separate processes for AI processing and file conversion tasks

Data Storage Environment: Separated for scalability and performance:

- PostgreSQL Database: The relational database management system for storing structured data.
- PostgreSQL Database: Primary relational database for structured data
- Elasticsearch Cluster: Distributed search engine for document indexing and retrieval
- Object Storage (S3-compatible): For PDF document storage with CDN integration for fast global downloads

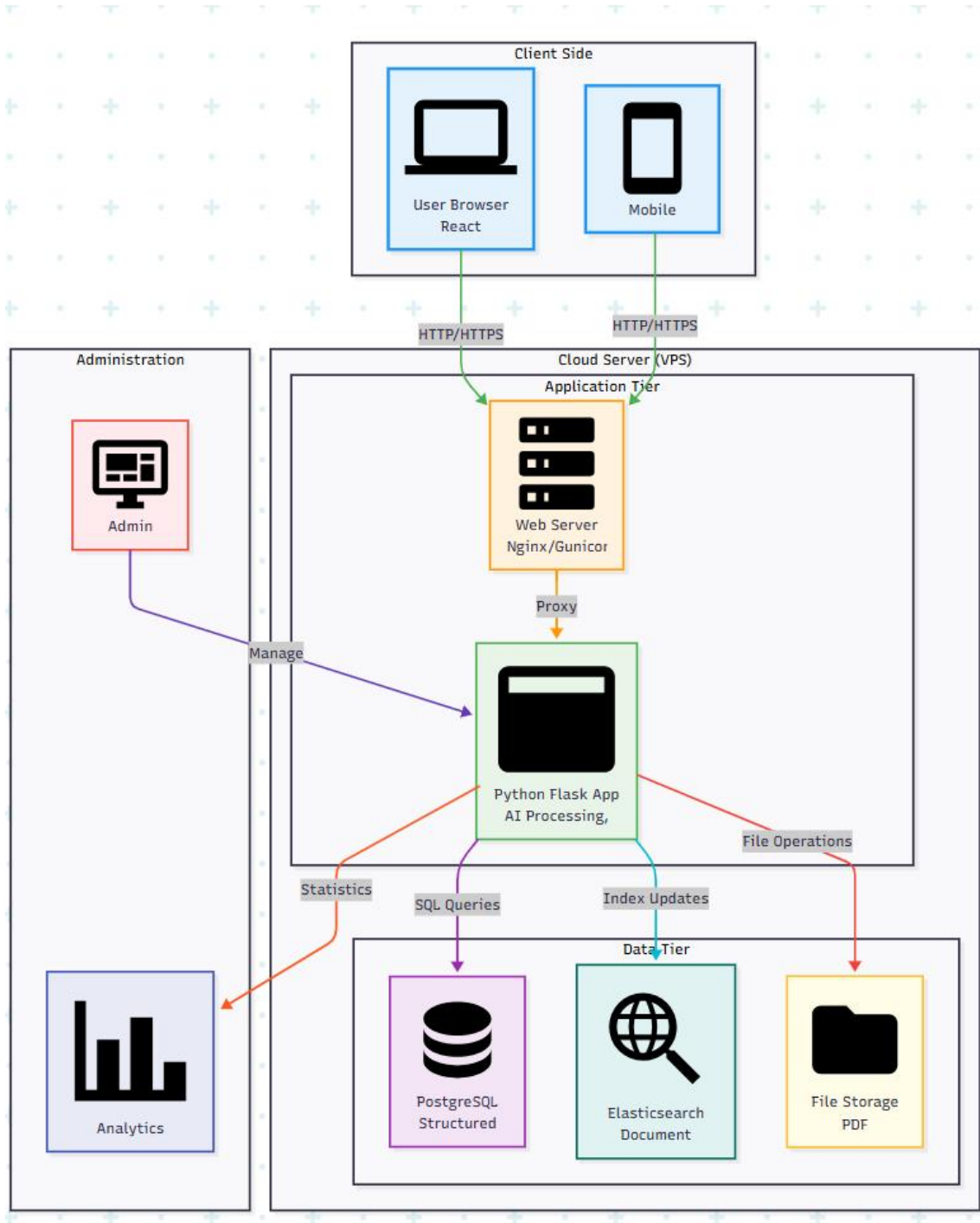


Figure 3.1: System Deployment Diagram

3.5. Architectural Design

To achieve a modular, scalable, and maintainable system, the software will be designed using a classic 3-Tier Architecture. This architectural pattern separates the application

into three logical and physical computing tiers. This separation of concerns makes the system easier to develop, test, and update.

The three tiers are:

1. **Presentation Tier (Frontend):**

This is the user interface and the outermost layer of the application. It is what the end-user directly interacts with. Its primary responsibility is to display information to the user and capture their input.

Technology: It will be built using standard web technologies: HTML, CSS, and JavaScript.

Function: It will render the search pages, dashboards, and profile pages. It does not contain any business logic; instead, it communicates with the application tier via a set of well-defined API (Application Programming Interface) calls over HTTP.

2. **Application Tier (Backend / Business Logic):**

This tier is the core of the system; it acts as the "brain" of the application. It is responsible for processing all the business logic and coordinating the application's functions.

Technology: It will be developed in Python, using a web framework such as Flask or Django.

Function: Its responsibilities are extensive:

- Handling user authentication and session management.
- Processing all incoming API requests from the presentation tier.
- Executing the document processing pipeline (PDF text extraction, data cleaning).
- Implementing the AI/ML algorithms (TF-IDF, LDA, Cosine Similarity).

- Coordinating data storage and retrieval by sending queries to the data tier.

3. Data Tier (Storage):

This tier is responsible for the persistent storage and management of all data. A key design decision for this project is to use a hybrid data storage approach, leveraging two specialized technologies for their respective strengths.

Technology 1: PostgreSQL Database (for Structured Data): A powerful open-source relational database (RDBMS).

Function: It will store all structured, relational data. This includes the Users, Institutions, Authors, and SavedSearches tables. It will also store the core metadata for Documents (like title, publication date, and file path) and the relationships between tables (e.g., which document belongs to which institution). Using an RDBMS ensures data integrity and consistency through the use of schemas and foreign keys.

Technology 2: Elasticsearch (for Unstructured Text Data): A specialized search and analytics engine.

Function: The full, extracted text from every research paper will be sent to Elasticsearch. Its sole purpose is to index this vast amount of unstructured text to provide extremely fast, powerful, and complex search capabilities (including semantic search and synonym support). Relying on Elasticsearch for search is vastly more efficient and scalable than performing full-text searches on a traditional relational database.

3.6. User Interface (UI) Design

The User Interface (UI) is the primary point of interaction for the end-user. The design philosophy is centered around creating a clean, intuitive, and efficient experience that empowers researchers without a steep learning curve. The UI design will be guided by the principles of clarity and focus, ensuring that the most important functionalities are easily accessible. The visual design, as shown in the prototypes in Section 2.4.6, will

utilize a professional and modern aesthetic with a consistent color scheme and typography.

The key pages and their design components are outlined below:

Search Page: This is the main entry point for discovery. It will feature a prominent search bar as the primary call to action. Below the search bar, advanced filtering options for Institution, Author, and Publication Date Range will be clearly laid out. This design allows for both simple keyword searches and more complex, faceted queries to be initiated with ease.

Search Results Page: This page is designed for clarity and scannability. Each result will be displayed in a well-defined card, presenting the document's title, authors, institution, and publication date. A "View Details" link will navigate the user to the full document page. The filters used for the search will be clearly displayed, and users will have the option to modify or clear them to perform a new search.

Researcher Dashboard: This is the personalized hub for each registered user. The design will focus on delivering relevant information at a glance. It will feature two main sections:

Saved Searches: A list of the user's saved queries, each with a notification count indicating the number of new matching documents that have been added.

Alerts: A chronological feed of new documents that match any of the user's saved searches, providing direct links to these new discoveries.

Profile Pages (Author & Institution): These pages will act as centralized information hubs. They will have a consistent layout, featuring the name of the author or institution at the top, followed by a list of all associated research documents available in the system. This design allows users to easily explore the entire body of work related to a specific person or organization.

3.7. Database Design

The database is the foundation of the system, responsible for storing all structured data in a reliable and consistent manner. The design is a direct implementation of the Class Diagram specified in Section 2.4.5. A relational database model using PostgreSQL was chosen to enforce data integrity through schemas, primary keys, and foreign key constraints.

The database will consist of the following primary tables:

Name	Data Type	Constraints	Description
userId	SERIAL	PRIMARY KEY	Unique identifier for the user.
name	VARCHAR(255)	NOT NULL	Full name of the user.
email	VARCHAR(255)	NOT NULL, UNIQUE	User's email, used for login.
passwordHash	VARCHAR(255)	NOT NULL	The securely hashed user password.
role	VARCHAR(50)	NOT NULL	User's role (e.g., 'Researcher', 'Administrator').

Table 3.1: Users Table Schema

Name	Data Type	Constraints	Description
institutionId	SERIAL	PRIMARY KEY	Unique identifier for the institution.
name	VARCHAR(255)	NOT NULL, UNIQUE	Official name of the institution.
description	TEXT		A brief description of the institution.

Table 3.2: Institutions Table Schema

Name	Data Type	Constraints	Description
authorId	SERIAL	PRIMARY KEY	Unique identifier for the author.

name	VARCHAR(255)	NOT NULL, UNIQUE	Full name of the author.
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Table 3.3: Authors Table Schema

Name	Data Type	Constraints	Description
docId	SERIAL	PRIMARY KEY	Unique identifier for the document.
title	TEXT	NOT NULL	The title of the research paper.
abstract	TEXT		The abstract of the paper.
publicationDate	DATE		The date of publication.
filePath	VARCHAR(255)	NOT NULL	Server path to the stored PDF file.
uploaderId	INTEGER	FOREIGN KEY (Users.userId)	ID of the user who uploaded it.
institutionId	INTEGER	FOREIGN KEY (Institutions.institutionId)	The affiliated institution.
downloadCount	INTEGER	DEFAULT 0	Number of times downloaded
status	VARCHAR(20)	DEFAULT 'Pending'	Moderation status ('Pending', 'Approved', 'Rejected')

Table 3.4: Documents Table Schema

Name	Data Type	Constraints	Description
docId	INTEGER	FOREIGN KEY (Documents.docId), PRIMARY KEY	Part of the composite primary key.
authorId	INTEGER	FOREIGN KEY (Authors.authorId), PRIMARY KEY	Part of the composite primary key.

Table 3.5: Document_Authors Table Schema (Linking Table)

Name	Data Type	Constraints	Description
searchId	SERIAL	PRIMARY KEY	Unique identifier for the saved search.
userId	INTEGER	FOREIGN KEY (Users.userId)	The user who saved the search.
searchName	VARCHAR(255)	NOT NULL	A user-defined name for the search.
queryParams	JSONB	NOT NULL	The search parameters

Table 3.6: SavedSearches Table Schema

Name	Data Type	Constraints	Description
queueId	SERIAL	PRIMARY KEY	Unique identifier for the moderation entry
docId	INTEGER	FOREIGN KEY (Documents.docId)	The document to be moderated
moderatorId	INTEGER	FOREIGN KEY (Users.userId)	The moderator assigned
status	VARCHAR(20)	DEFAULT 'Pending'	Current status ('Pending', 'Reviewed')
notes	TEXT		Moderator's comments

Table 3.7: ModerationQueue Table Schema

Name	Data Type	Constraints	Description
logId	SERIAL	PRIMARY KEY	Unique identifier for the download log

Name	Data Type	Constraints	Description
docId	INTEGER	FOREIGN KEY (Documents.docId)	The downloaded document
downloadDate	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of download
ipAddress	VARCHAR(45)		IP address of downloader

Table 3.8: DownloadLogs Table Schema

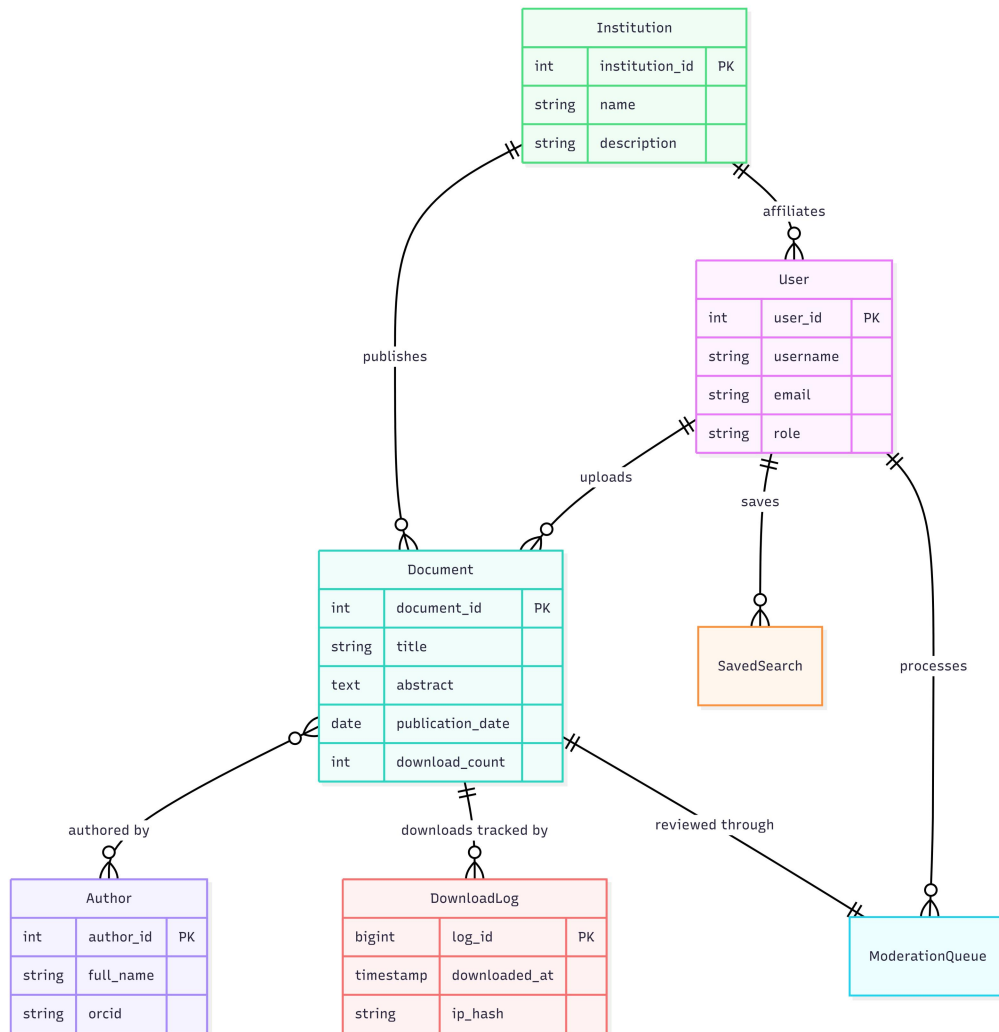


Figure 3.2: Entity-Relationship Diagram (ERD)

3.8. Machine Learning Algorithm Design

The core intelligence of the system lies in its ability to automatically process and understand the content of uploaded research documents. This is achieved through a well-defined pipeline of Natural Language Processing (NLP) and machine learning techniques. The pipeline is designed to transform unstructured text from a PDF into structured, searchable, and analyzable data.

The pipeline consists of the following sequential steps, which are executed every time a new document is uploaded:

1. PDF Text Extraction:

- **Purpose:** To extract the raw textual content from the uploaded PDF file.
- **Tool:** A Python library such as PyMuPDF or pdfplumber will be used.
- **Process:** The library will open the PDF file and iterate through its pages, extracting all text blocks. The extracted text from all pages will be concatenated into a single string for further processing.

2. Text Preprocessing and Cleaning:

- **Purpose:** To clean the raw text and prepare it for accurate machine learning analysis. This step is crucial for reducing noise and improving the quality of the results.
- **Process:** The following sub-steps will be performed:
 - **Lowercasing:** All text will be converted to lowercase to ensure uniformity (e.g., "Health" and "health" are treated as the same word).
 - **Punctuation Removal:** All punctuation marks (e.g., commas, periods, hyphens) will be removed.

- **Stopword Removal:** Common, low-value words (known as stopwords, e.g., 'the', 'is', 'a', 'an', 'in') will be removed from the text. A standard English stopwords list from a library like NLTK or spaCy will be used.

3. **Keyword and Tag Generation (TF-IDF):**

- **Purpose:** To automatically identify the most significant keywords or "tags" for a document.
- **Algorithm:** Term Frequency-Inverse Document Frequency (TF-IDF).
- **Process:** TF-IDF will be used to calculate a score for each word in the document. This score reflects how important a word is to that specific document in the context of the entire collection of documents. Words with the highest TF-IDF scores (e.g., "malaria," "placenta," "kinetics") will be selected and stored as the document's primary keywords. These keywords are also used as the basis for the recommendation system.

4. **Topic Modeling (LDA):**

- **Purpose:** To automatically categorize each document into a set of high-level topics.
- **Algorithm:** Latent Dirichlet Allocation (LDA).
- **Process:** The LDA model will be trained on the entire corpus of preprocessed text from all documents. It will analyze the patterns of co-occurring words to discover a set of abstract "topics" (e.g., one topic might be characterized by words like "virus," "vaccine," "infection," while another might be "data," "model," "statistics"). Each document is then assigned a probability distribution across these topics, allowing the system to classify it under its most likely category.

5. **Content-Based Recommendation (Cosine Similarity)**

- **Purpose:** To recommend other relevant documents to a user who is viewing a specific paper.
- **Algorithm:** Cosine Similarity.

- **Process:** The recommendation engine will use the TF-IDF vectors generated in Step 3. Each document is represented as a vector in a high-dimensional space. To find documents similar to a given document, the system will calculate the cosine of the angle between its vector and the vectors of all other documents in the repository. A cosine value close to 1 indicates a very high similarity. The top N documents with the highest similarity scores will be presented to the user as "Related Articles."

3.9. Summary

This chapter has provided a comprehensive technical blueprint for the design of the Intelligent Knowledge Management System. It began by establishing the high-level architectural goals and defining a scalable 3-Tier Architecture built upon a robust cloud deployment model. The design of the User Interface was detailed, aligning with the prototypes from Chapter 2 to ensure a user-centric experience.

A crucial part of this chapter was the detailed Database Design, which specified the schema for a PostgreSQL database, ensuring data integrity and providing a solid foundation for all system functionalities. Finally, the Machine Learning Algorithm Design detailed the five-step pipeline that will power the system's intelligence, from raw text extraction to automated topic modeling and content-based recommendations.